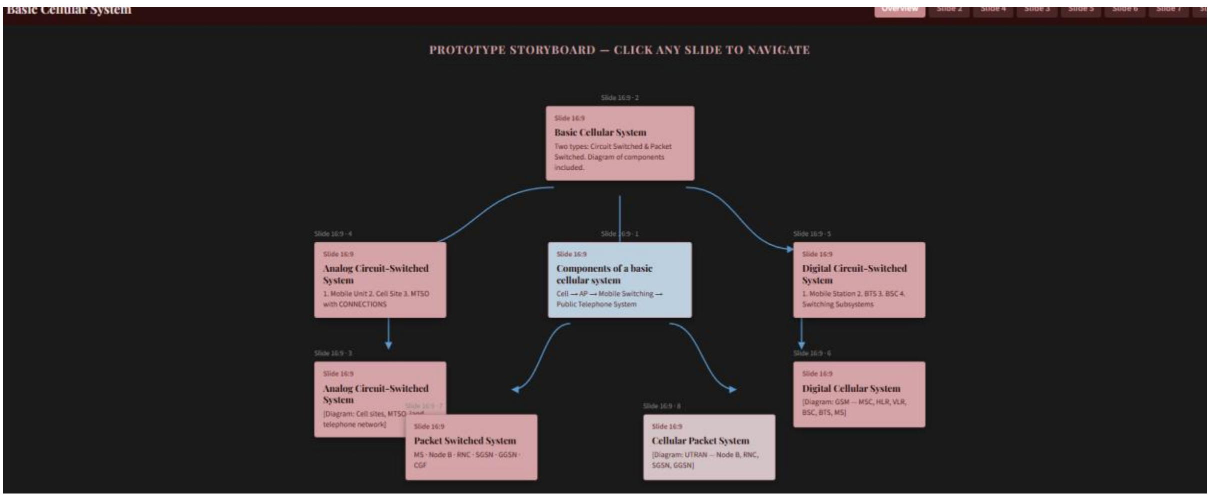


Basic Cellular System

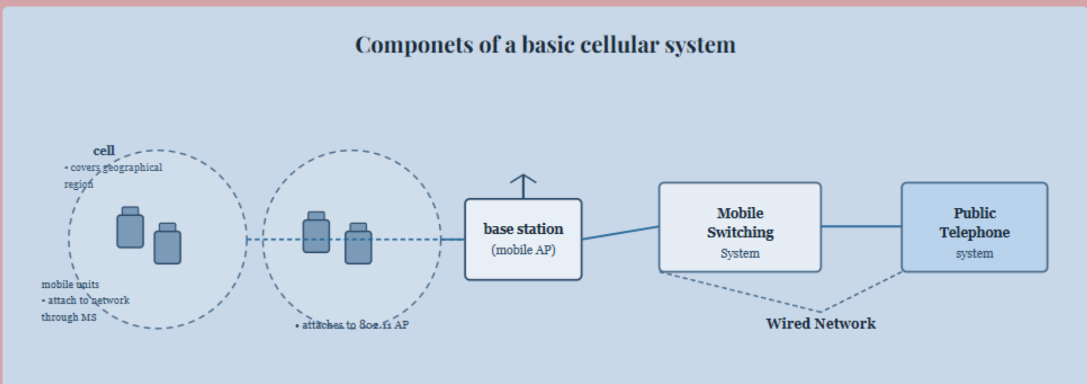
There are mainly two types of Basic Cellular System:

1.Circuit Switched :

In a circuit-switched system, each traffic channel is dedicated to a user until its cell is terminated.

2.Packet Switched:

In packet-switching, the packets are sent towards the destination irrespective of each other. There is no predetermined path; the decision as to which node to hop to in the next step is taken only when a node is reached. Each packet finds its way using the information it carries, such as the source and destination IP addresses.



Analog Circuit-Switched System

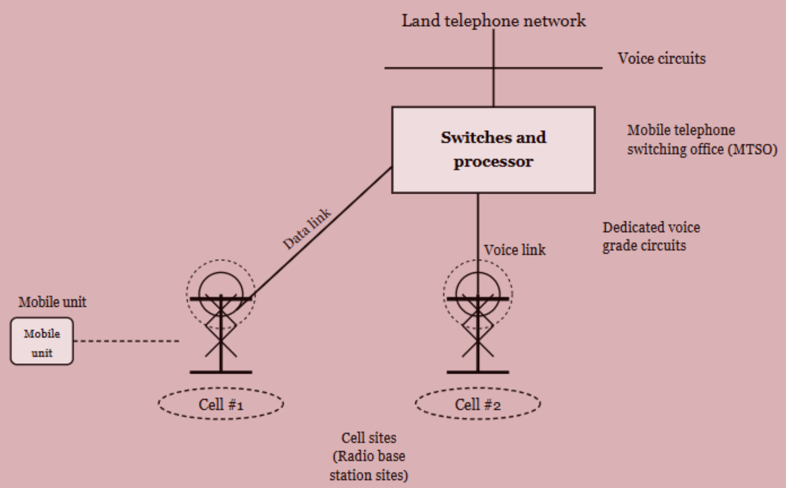
A basic analog cellular system consists of three subsystems:

1. A Mobile Unit

2. A Cell Site

3. A Mobile Telephone Switching Office (MTSO) with **<u>CONNECTIONS</u>** to link the three subsystems

Analog Circuit-Switched System



Analog cellular system

Digital Circuit-Switched System

A Basic Digital System consists of four elements:

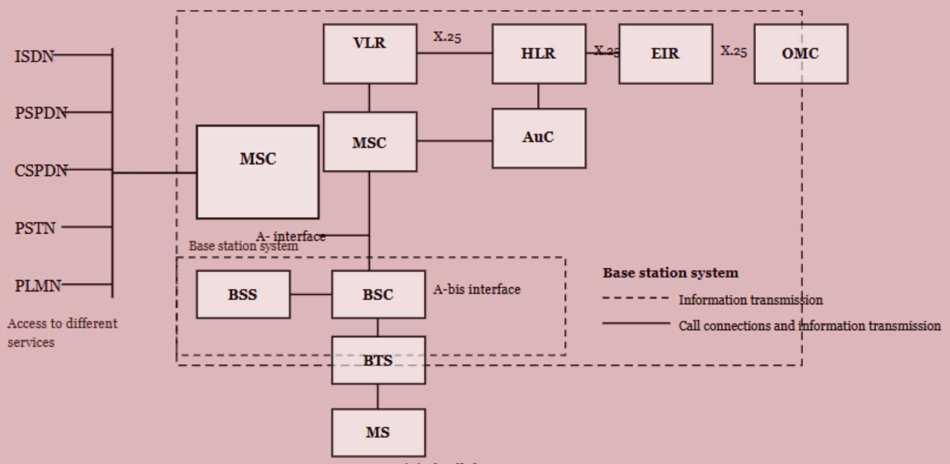
1. Mobile Station

2. Base Transceiver Station (BTS)

3. Base Station Controller (BSC)

4. Switching Subsystems

Digital Circuit-Switched System



Digital cellular system

Packet Switched System

A cellular packet-switched system has six elements as follows:

1. MS (Mobile Station)

2. Node B

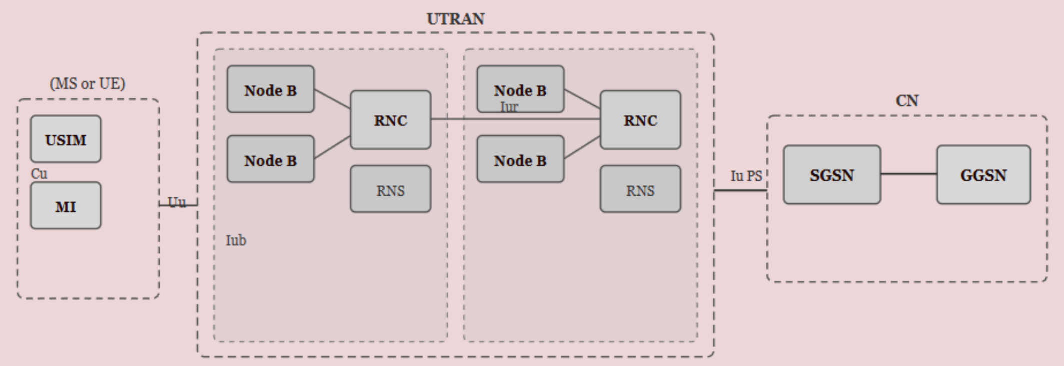
3. RNC (Radio Network Controller)

4. SGSN (Service GPRS Support Node)

5. GGSN (Gateway GPRS Support Node)

6. CGF (Changing Gateway Function)

Cellular Packet System



Cellular packet system